

# The One-Baryon Universe: Hyperon Identity and Strangeness Topology

Version 2.0 — Revised & Scientifically Deepened | June 2026

**Classification:** Tardigradia Game Engine | Physics Deep Dive Series | TGPU v2.0 **Particle Class:** Baryon (Fermion, Composite) | Quark content: uds (Lambda) | Spin: 1/2 | Strangeness: -1 **Source:** Extends v1.0Baryon Deep Dive\_ One-Particle Universe.pdf (V1.0)

## Δ Change Log V1.0 → V2.0

| Parameter                            | V1.0                                       | V2.0                                                                                                                                                                                                        | Source                 |
|--------------------------------------|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| <b>Lambda baryon lifetime</b>        | $\tau = 2.63 \times 10^{-10}$ s (PDG 2020) | $\tau = 2.631 \pm 0.020 \times 10^{-10}$ s (PDG 2024, 0.8% precision improvement)                                                                                                                           | PDG 2024               |
| <b>Strangeness production at LHC</b> | General discussion                         | LHC Run 3 (2022-2025): ALICE strangeness enhancement in pp at 13.6 TeV — unexpectedly high Lambda production in high-multiplicity events (QGP precursor?)                                                   | ALICE Run 3 2023       |
| <b>Lattice QCD hyperon masses</b>    | Phenomenological mass estimates            | V2.0: BMW lattice QCD 2024 — Lambda mass 1115.83 MeV (measured: 1115.683 MeV), systematic error 0.15%. Validates quark model parameters                                                                     | BMW Collaboration 2024 |
| <b>GPU strangeness topology</b>      | Conceptual data structure                  | V2.0: Strange quark topological charge tracked as $S \in \{-3, -2, -1, 0, +1, +2, +3\}$ INTEGER. Bit-packed in 4 bits of particle state vector. Preserves baryon number $B=+1$ constraint in WebGPU compute | OEU V2.0 engine spec   |

## 1. The One-Particle Universe Framework — V2.0 Update

The foundational architecture established in V1.0 — Worldline Monism, the “Origami Universe” metaphor, and the Object-Oriented ontology distinguishing intrinsic from extrinsic properties —

is **fully preserved** and remains physically rigorous. V2.0 layers NEW discoveries from 2022-2026 atop this framework.

### 1.1 WebGPU Compute Architecture (V2.0 Core Infrastructure)

V1.0 designed for WebGL/Three.js on the CPU-side particle update loop. V2.0 upgrades to WebGPU (available in Chrome 113+ May 2023, Firefox 2024):

```
// WGSL Compute Shader - V2.0 Particle Integration
@binding(0) @group(0) var<storage, read_write> particles: array<ParticleState>;

struct ParticleState {
    position: vec4f,
    momentum: vec4f,
    spin: vec2f,           // Spinor components
    berry_phase: f32,      // Geometric phase accumulated
    proper_time: f32,      // tau - worldline parameter
    color_charge: u32,      // Packed color in 2 bits (R=0,G=1,B=2)
    quantum_numbers: u32,  // Packed: isospin, strangeness, charm, beauty
    decay_lifetime: f32,
    entanglement_id: i32, // -1 = unentangled; >=0 = entanglement group
}

@compute @workgroup_size(64)
fn updateParticles(@builtin(global_invocation_id) id: vec3u) {
    let i = id.x;
    if (i >= arrayLength(&particles)) { return; }

    var p = particles[i];

    // Feynman-Schwinger proper time evolution
    let dtau = uniforms.dt / p.mass; // proper time step

    // Update worldline parameter
    p.proper_time += dtau;

    // Berry phase accumulation (topological phase)
    p.berry_phase += dot(uniforms.gauge_field, p.momentum.xyz) * dtau;

    // Lorentz force (for charged particles)
    let F = cross(uniforms.B_field, p.momentum.xyz) * p.charge_sign;
    p.momentum.xyz += F * uniforms.dt;
    p.position.xyz += p.momentum.xyz * uniforms.dt / p.energy;

    particles[i] = p;
}
```

### 1.2 Updated Physical Constants (CODATA 2022)

| Constant | V1.0 Value                                           | V2.0 Value (CODATA 2022)                                     |
|----------|------------------------------------------------------|--------------------------------------------------------------|
| $m_e$    | $9.10938356 \times 10^{-31} \text{ kg}$              | $9.1093837139 \times 10^{-31} \text{ kg}$                    |
| $e$      | $1.60217663 \times 10^{-19} \text{ C}$               | $1.602176634 \times 10^{-19} \text{ C (exact)}$              |
| $\hbar$  | $1.054571817 \times 10^{-34} \text{ J}\cdot\text{s}$ | $1.054571817 \times 10^{-34} \text{ J}\cdot\text{s (exact)}$ |

| Constant | V1.0 Value                     | V2.0 Value (CODATA 2022)               |
|----------|--------------------------------|----------------------------------------|
| c        | 2.99792458×10 <sup>8</sup> m/s | 2.99792458×10 <sup>8</sup> m/s (exact) |
| alpha    | 1/137.036                      | 1/137.035999084 ±0.000000021           |

## 2. V2.0 Enhanced Data Structures

Extending V1.0 struct definitions with 2022-2026 discoveries:

```
// V2.0 Universal Particle State – extends V1.0
struct ParticleInstanceV2 {
    // Inherited from V1.0
    double proper_time;           // tau worldline parameter (meter)
    glm::dvec3 position;          // 3-space vector
    glm::dvec3 momentum;          // 3-momentum (relativistic)
    float spin_z;                 // +0.5 or -0.5 (or eigenvalue)
    float berry_phase;            // Accumulated Berry/Aharonov-Bohm phase

    // V2.0 ADDITIONS
    float entanglement_r;         // Reduced density matrix purity 0-1
    int entanglement_partner_id;  // -1 = none; global particle ID
    float flavor_oscillation_phase; // For neutrinos/kaons
    uint32_t color_bits;          // 3-bit: R(b0),G(b1),B(b2)
    float decay_probability_dt;   // P(decay in dt) for unstable particles
    float vacuum_alignment;       // Order parameter for condensate mode

    // WebGPU Buffer packing
    // Total: ~100 bytes/particle * 10^6 particles = 100 MB GPU memory
    float _pad[2];                // Align to 16 bytes for WGS
};
```

## 8. V2.0 Benevolence Metric Update

The “Benevolence” concept central to all OEU documents is given a precise V2.0 mathematical definition grounded in 2022-2026 quantum information theory:

**Benevolence = Quantum Coherence + Information Preservation + Symmetry Maintenance**

$$B(t) = C(t) \cdot F(t) \cdot S(t)$$

Where: - C(t) = l<sub>1</sub>-norm coherence of particle density matrix (Baumgratz et al. 2014) - F(t) = quantum Fisher information / classical Fisher information (ratio ≥ 1) - S(t) = fraction of symmetries unbroken at time t

**V2.0 game mechanic:** Benevolence score increases when: - Particles maintain long entanglement coherence (high C) - Topological protections are active (Berry phase winding conserved) - Symmetry groups are preserved in interactions (no spurious symmetry breaking)

*End of v1.0Baryon Deep Dive\_ One-Particle Universe V2.0 | Tardigradia Game Engine SubParticles Series This document supersedes V1.0 in all physics specifications.*